



NORTHERN BEACHES SECONDARY COLLEGE

MANLY SELECTIVE CAMPUS

HIGHER SCHOOL CERTIFICATE

Trial Examination

2017

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3hours
- Use **black** pen
- Write your Student Number at the top of each page
- Section I – Multiple Choice – use the Answer Sheet provided
- Section II – Free Response – use a separate booklet for **each** question.
- Board approved calculators and templates may be used.
- Reference sheet provided.

Section I Multiple Choice

- 10 marks
- Attempt all questions

Section II – Free Response

- Questions 11-16 – 90 marks
- Each question is of equal value
- All necessary working should be shown in every question.

Weighting: 40%

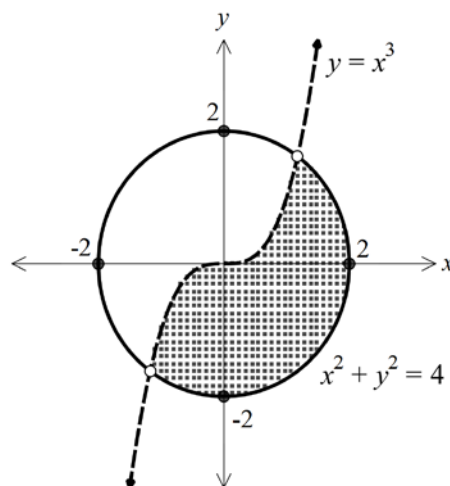
Section 1 Multiple Choice: Attempt Questions 1 – 10

Answer questions on the provided answer sheet.
Allow approximately 15 minutes for this section.

Q. 1 36.1984 written in scientific notation, correct to 4 significant figures is:

- (A) 3.620×10
- (B) 3.62×10
- (C) 3.620×10^{-1}
- (D) 3.6198×10^{-1}

Q. 2 The inequalities which define the shaded region shown in the diagram are:



- (A) $x^2 + y^2 \geq 4$ and $y < x^3$
- (B) $x^2 + y^2 \geq 4$ and $y \geq x^3$
- (C) $x^2 + y^2 \leq 4$ and $y > x^3$
- (D) $x^2 + y^2 \leq 4$ and $y < x^3$

Q.3 A line in the form of $px + qy = 5$ passes through the points (0, 5) and (2, 1). The gradient (m) and y -intercept (b) are:

(A) $m = 1$ $b = 2$

(B) $m = -2$ $b = 5$

(C) $m = 5$ $b = -2$

(D) $m = 2$ $b = 1$

Q. 4 The fourth, fifth and sixth terms of an arithmetic series are -3 , 5 and 13 respectively. The first term of this series is:

(A) 21

(B) -11

(C) -27

(D) 8

Q. 5 Given $\log \frac{a}{b} + \log \frac{b}{a} = \log (a + b)$, then the correct statement below is:

(A) $a + b = 1$

(B) $a - b = 1$

(C) $a = b$

(D) $a^2 - b^2 = 1$

Q. 6 The function $f(x)$ is defined by $f(x) = \begin{cases} \frac{4}{x} & : x > 1 \\ 4^x & : x \leq 1 \end{cases}$.

What is the value of $f(0.5) + f(2)$?

- (A) 4
- (B) 10
- (C) 18
- (D) 24

Q. 7 $\int \frac{x-4}{x^2} dx$ can be expressed as:

- (A) $\frac{1}{2} \ln x^2 + 4x + C$
- (B) $\ln x^2 + \frac{4}{x} + C$
- (C) $\ln x - 4x + C$
- (D) $\ln x + \frac{4}{x} + C$

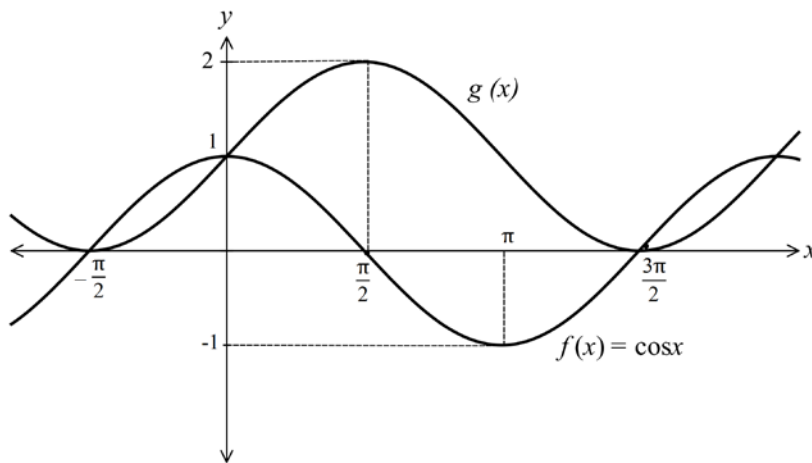
Q. 8 Which of the following statements is true if $(2, -5)$ is a minimum turning point of $f(x)$ and $f(x) = -f(-x)$?

- (A) $(-2, -5)$ is a maximum turning point of $f(x)$
- (B) $(-2, 5)$ is a minimum turning point of $f(x)$
- (C) $(-2, -5)$ is a minimum turning point of $f(x)$
- (D) $(-2, 5)$ is a maximum turning point of $f(x)$

- Q. 9 The quadratic equation $x^2 - 9x + 16 = 0$ has roots α and β .
The value of $\sqrt{\alpha} + \sqrt{\beta}$ is:

- (A) 3
(B) $\sqrt{13}$
(C) $\sqrt{17}$
(D) 17

- Q. 10 The diagram shows the graphs of $f(x) = \cos x$ and $g(x)$.



The equation of $g(x)$ is:

- (A) $g(x) = f\left(x - \frac{\pi}{2}\right) + 2$
(B) $g(x) = f\left(x + \frac{\pi}{2}\right) + 1$
(C) $g(x) = f\left(x - \frac{\pi}{2}\right) + 1$
(D) $g(x) = f\left(x + \frac{\pi}{2}\right) + 2$

End of Multiple Choice

Section II 90 marks**90 marks****Attempt Questions 11–16****Allow about 2 hours and 45 minutes for this section**

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations

Question 11: Start A New Booklet**15 Marks**

- a. Fully factorise $16 - 2x^3$. 2
- b. Solve $|2x - 5| < 1$. 2
- c. Evaluate $\lim_{x \rightarrow -1} \frac{2x^2 + 3x + 1}{x + 1}$ 2
- d. Find $\int_0^{\frac{\pi}{3}} \cos\left(\frac{x}{2}\right) dx$ 2
- e. Find the values of p and q for which $p + q\sqrt{6} = \frac{12\sqrt{6}}{\sqrt{6} - 2}$. 2
- f. Sketch the graph of $(x - 2)^2 + y^2 = 4$. 2
- g. Show that $\frac{d}{dx} \left[2x(x - 4)^3 \right] = 8(x - 1)(x - 4)^2$. 3

End of Question 11

Question 12 Start a New Booklet**15 Marks**

- a. The vertices of $\triangle ABC$ are A(1, 8) B(7, 4) and C(1,0) as shown below, where AQ is the altitude.

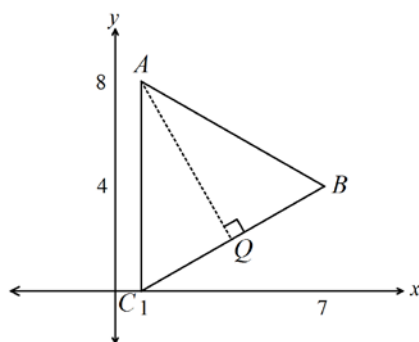


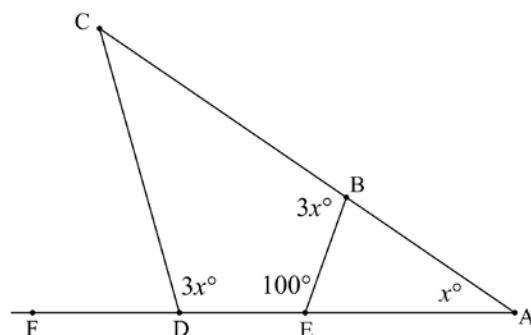
Diagram not to scale

- i. Show the gradient of AQ is $-\frac{3}{2}$ 1
 - ii. Hence, determine the equation of the altitude AQ in general form. 1
 - iii. Hence determine the length of BQ.
(Leave your answer in exact form) 2
- b. In a geometric series, the fifth term is $\frac{1}{9}$ and the eight term is $\frac{1}{243}$.
Determine the value of the common ratio (r). 2
- c. What is the value of $\sum_{m=1}^{\infty} 5\left(\frac{2}{5}\right)^{m-1}$? 2
- d. Find $\int \frac{1}{(2x+3)^5} dx$. 2

Question 12 continues on the next page.

Question 12 continued:

e.



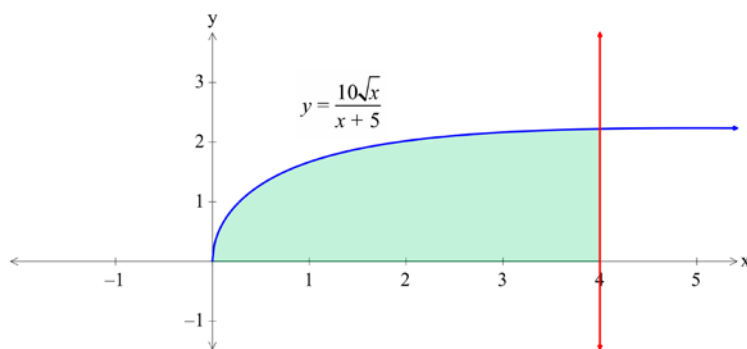
The points B and E lie on the sides AC and AD respectively of $\triangle ACD$.
The point F lies on AD produced, as shown in the diagram.

(i) Copy the diagram into your answer booklet.
Make your diagram one third of a page.

(ii) Find the value of x , giving reasons.

2

f The area bounded by the curve $f(x) = \frac{10\sqrt{x}}{x+5}$, the x -axis and the line $x = 4$ is shown in the diagram below.



i. Using the Trapezoidal Rule with 5 functional values, determine the approximate area of the shaded region.
Give your answer correct to two decimal places.

2

ii. Is the estimate calculated in part i) greater or less than the exact area?
Give a reason to justify your answer.

1

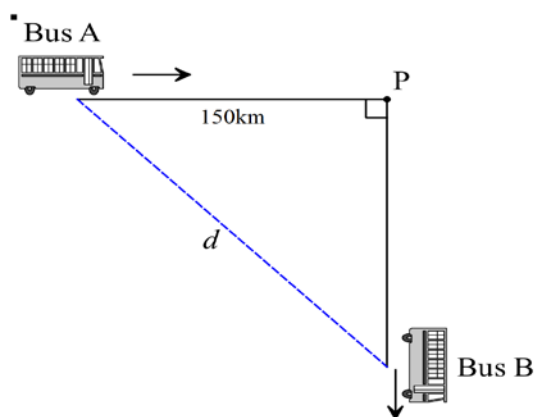
End of Question 12

Question 13 Start A New Booklet**15 Marks**

a. Find $\int (1 + \tan x) \, dx$.

2

- b. Two school buses are travelling along straight roads which intersect at right angles at the point P, as shown in the diagram.



Initially, Bus A is 150km due west of P and is travelling towards P at 50km/hr. At the same time Bus B leaves P and travels due south at 40km/hr. Let d km be the distance between Bus A and Bus B at t hours after the buses start moving.

- i. Show that Bus A is $(150 - 50t)$ km from P after t hours

1

- ii. Show that

2

$$d = \sqrt{4100t^2 - 15000t + 22500} ,$$

where d is the distance between the buses when $0 < t < 3$.

- ii. Find the value of t which gives the minimum value of d .
Answer correct to one decimal place.

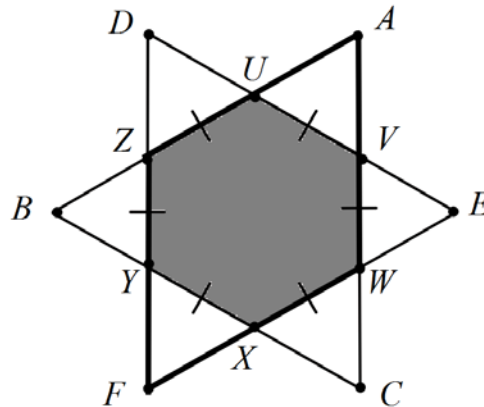
2

Question 13 continues on the next page.

Question 13 continued:

- c. Find the value of the derivative of $y = 4\tan(2x) - \frac{4x^2}{\pi}$ when $x = \frac{\pi}{12}$ 2

- d. The diagram shows a six pointed star which is drawn using two triangles, $\triangle ABC$ and $\triangle FDE$. The intersection of the two triangles is a regular hexagon.



- i. Show that $\triangle AVU$ is an equilateral triangle. 2
- ii. Similarly $\triangle VEW$, $\triangle WCX$, $\triangle XFY$, $\triangle YBZ$ and $\triangle ZDU$ are all equilateral triangles. 2

Prove that $ZAWF$ is a rhombus.

- e. Find, in simplest form, $\frac{d}{dx} \left(\frac{\cos x}{1 - \sin x} \right)$. 2

End of Question 13

Question 14 Start A New Booklet**15 Marks**

- a. Sketch the graph of $y = \ln(x - 4)$, clearly indicating the x -intercept and any asymptotes. **2**
- b. Consider the curve with equation $y = 4x^4 + 8x^3 + 1$.
- Find the stationary points and determine their nature. **4**
 - Show that the point $(-1, -3)$ is a point of inflexion. **2**
 - Sketch the curve, labelling the stationary points, points of inflexion and y -intercept. **3**
- c. A particle moves in a straight line such that its displacement x in metres is given by
- $$x = 70e^{-\frac{t}{10}} - 20t, \text{ where } t \text{ is time in seconds.}$$
- Find the initial displacement of the particle. **1**
 - Will the particle ever come to rest?
Justify your answer using appropriate calculations. **2**
 - Find the distance travelled by the particle in the first 3 seconds.
Give the answer correct to 2 decimal places. **1**

End of Question 14

Question 15 Start A New Booklet**15 Marks**

- a. The curve $f(x)$ has a minimum turning point at $(2, -10)$. The second derivative is given by the equation $f''(x) = 12x - 10$. Determine the equation of $f(x)$. **3**

- b. Determine the value of k given $\int_{-2}^2 (x^7 - x + k) dx = 16$ where k is constant. **2**

- c. The amount of caffeine, $C(t)$, in milligrams in your system after drinking a cappuccino is given by

$$C(t) = 105e^{-kt},$$

where k is a constant and t is the time in hours that have passed since drinking the cappuccino.

- (i) After one hour the caffeine in your system has decreased by 40%. Find the exact value of k . **2**
- (ii) When will there be 10 milligrams of caffeine remaining in your system? Give the answer correct to 2 significant figures. **1**
- d. A quadratic function is defined by $f(x) = x^2 - 2kx + (2k + 3)$.
- i) Find the values of k for which the equation $f(x) = 0$ has two real roots. **2**
- ii) Find the values of k for which the solutions to $f(x) = 0$ are both positive. **2**

- d. The rate at which the height of a Jacaranda tree grows is given by

$$\frac{dh}{dt} = \frac{110}{(t + 4)^2} \text{ metres per year,}$$

where h is the height of the tree in metres and t is the number of years that have passed since the tree was an established seedling with a height 0.5 m.

- Find the height of the tree when $t = 5$, correct to 1 decimal place. **3**

End of Question 15

Question 16 Start A New Booklet**15 Marks**

- a. Penny borrows \$250 000 to be repaid at a reducible interest rate of 0.4% per month. Let A_n be the amount owing at the end of n months and M be the monthly repayment.

(i) Show that $A_2 = 250\,000(1.004)^2 - M(1 + 1.004)$ 2

(ii) Show that $A_n = 250\,000(1.004)^n - M \left(\frac{(1.004)^n - 1}{0.004} \right)$ 1

(iii) If she repays the amount in 120 months, then show that $M = \$2627$, to the nearest dollar. 1

(iv) Penny decides to pay off the loan by making monthly payments of \$3500 instead of \$2627. 3

Show that Penny will make 84 repayments of \$3500 and then a final part payment of \$1001.64.

b. i Show $\frac{\sec\theta}{\operatorname{cosec}\theta} = \tan\theta$. 1

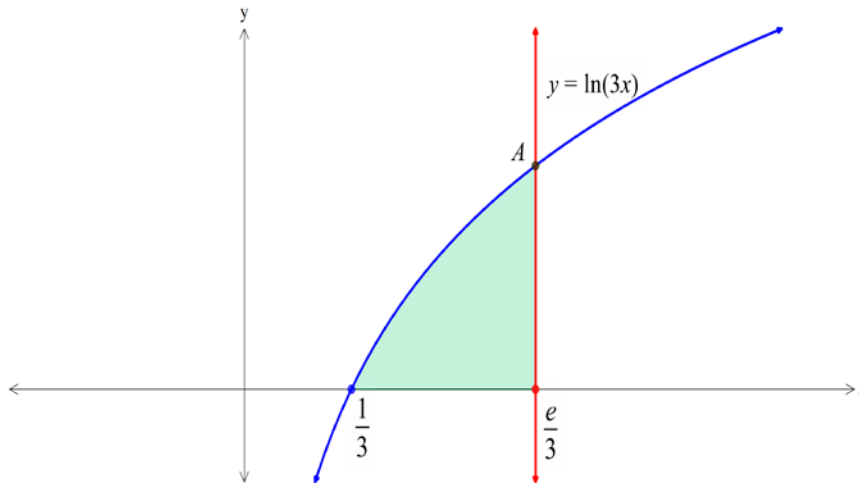
ii Hence, or otherwise, solve $3\sec^2\left(\frac{x}{2}\right) = \operatorname{cosec}^2\left(\frac{x}{2}\right)$ for $0 \leq x \leq 2\pi$. 3

Question 16 continues on the next page.

Question 16 continued:

- b. The shaded region shown below is rotated around the y-axis.

The point A is the intersection of the curve $y = \ln(3x)$ and the line $x = \frac{e}{3}$.



- | | | |
|----|---|---|
| i | Determine the y – value of point A. | 1 |
| ii | Determine the exact volume of the solid of revolution formed. | 3 |

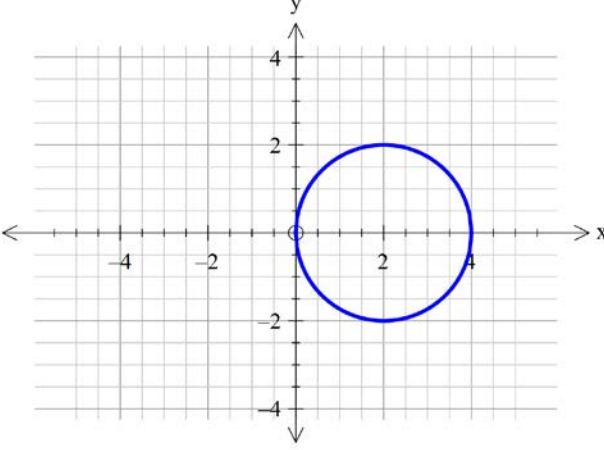
End of Examination

MSC HSC Trial Examination 2017- Solutions

Q1	A	$3.61984 \times 10^1 = 3.620 \times 10^1$ (4 sig)
Q2	D	Inside the circle and below but not including the cubic.
Q3	B	sub (0, 5) into $px + qy = 5$ $p(0) + q(5) = 5$ $5q = 5$ $q = 1$ $\therefore px + y = 5$ sub (2, 1) into $px + y = 5$ $p(2) + 1 = 5$ $p = 2$ $\therefore 2x + y = 5$ $m = -2$ $b = 5$
Q4	C	$T_4 = -3$ $T_5 = 5$ $T_6 = 13$ $\therefore d = 8$ $T_1 = T_4 - 3d$ $= -3 - 3(8)$ $= -27$
Q5	A	$\log\left(\frac{a}{b} * \frac{b}{a}\right) = \log(a + b)$ $a + b = 1$
Q6	A	$f(0.5) = 4^{\frac{1}{2}} = 2$ $f(2) = \frac{4}{2} = 2$ $f(0.5) + f(2) = 4$
Q7	D	$\int \frac{x}{x^2} - \frac{4}{x^2} dx = \int \frac{1}{x} - 4x^{-2} dx$ $= \ln x - \frac{4x^{-1}}{-1} + C$ $= \ln x + \frac{4}{x} + C$
Q8	D	$f(x) = -f(-x) \Rightarrow f(-x) = -f(x)$ $\therefore$ function is odd If minimum turning point at (2, -5) then maximum turning point (-2, 5)
Q9	C	

		$\alpha + \beta = 9$ $\alpha \times \beta = 16$ $(\sqrt{\alpha} + \sqrt{\beta})^2 = (\sqrt{\alpha})^2 + 2\sqrt{\alpha}\sqrt{\beta} + (\sqrt{\beta})^2$ $= \alpha + \beta + 2\sqrt{\alpha\beta}$ $= 9 + 2\sqrt{16}$ $= 9 + 8$ $= 17$ $\therefore \sqrt{\alpha} + \sqrt{\beta} = \sqrt{17}$
Q10	C	Curve has been lifted by 1 unit vertically and translated $\frac{\pi}{2}$ units to the right therefore $g(x) = f\left(x - \frac{\pi}{2}\right) + 1$

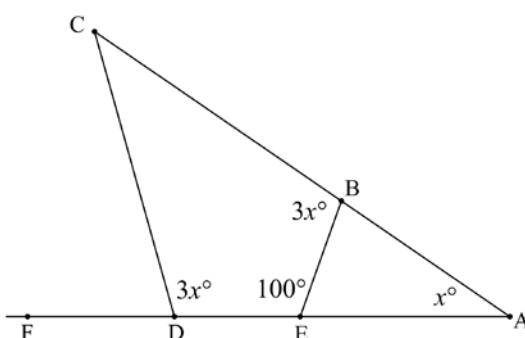
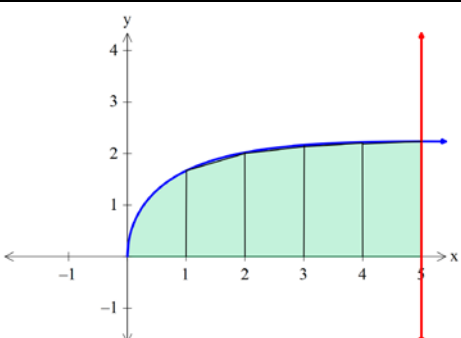
Q11a	$16 - 2x^3 = 2(8 - x^3)$ $= 2(2 - x)(4 + 2x + x^2)$	2 marks for correct solution from correct working 1 mark for taking out common factor
Q11b	$-1 < 2x - 5 < 1$ $4 < 2x < 6$ $\therefore 2 < x < 3$	2 marks for correct solution from correct working
Q11c	$\lim_{x \rightarrow -1} \frac{2x^2 + 3x + 1}{x + 1} = \lim_{x \rightarrow -1} \frac{(2x + 1)(x + 1)}{x + 1}$ $= \lim_{x \rightarrow -1} 2x + 1$ $= -2 + 1$ $= -1$	2 marks for correct solution from correct working
Q11d	$\int_0^{\frac{\pi}{3}} \cos\left(\frac{x}{2}\right) dx$ $= 2 \left[\sin \frac{x}{2} \right]_0^{\frac{\pi}{3}}$ $= 2 \left(\sin \frac{\pi}{6} - \sin 0 \right)$ $= 2 \times \frac{1}{2} = 1$	2 marks for correct solution from correct working
Q11e	$\frac{12\sqrt{6}}{\sqrt{6} - 2} = \frac{12\sqrt{6}}{\sqrt{6} - 2} \times \frac{\sqrt{6} + 2}{\sqrt{6} + 2}$ $= \frac{72 + 24\sqrt{6}}{6 - 4}$ $= \frac{72 + 24\sqrt{6}}{2}$ $= 36 + 12\sqrt{6}$ $\therefore p = 36 \text{ and } q = 12$	2 marks for correct solution from correct working

Q11f	 Centre (2,0) Radius=2	2 marks for correct graph
Q11g	$U = 2x \qquad V = (x - 4)^3$ $U' = 2 \qquad V' = 3(x - 4)^2$ $\frac{d}{dx}[2x(x - 4)^3] = VU' + UV'$ $= 2(x - 4)^3 + 6x(x - 4)^2$ $= 2(x - 4)^2[(x - 4) + 3x]$ $= 2(x - 4)^2(4x - 4)$ $= 8(x - 4)^2(x - 1)$	3 marks for correct solution from correct working

Markers Comments

Q12a(i)	$m_{AQ} \times m_{CB} = -1$ $m_{AQ} = -\frac{1}{m_{CB}}$ $m_{CB} = \frac{4-0}{7-1} = \frac{2}{3}$ $\therefore m_{AQ} = -\frac{3}{2}$	1 mark – correct demonstration from gradient of CB
Q12a(ii)	$y = mx + b$ $y = -\frac{3}{2}x + b$ $x = 1 \Rightarrow y = 8$ $8 = -\frac{3}{2} + b$ $b = 8 + \frac{3}{2} = \frac{19}{2}$ $y = -\frac{3}{2}x + 9.5$ $2y = -3x + 19$ $3x + 2y - 19 = 0$	1 mark – correct formula ie. Both gradient and y-intercept correct
a-iii	$D = \frac{ ax + by + c }{\sqrt{a^2 + b^2}}$ $(x,y) \Rightarrow (7,4)$ $3x + 2y - 19 = 0$ $D = \frac{ 3 \times 7 + 2 \times 4 - 19 }{\sqrt{9 + 4}}$ $= \frac{ 10 }{\sqrt{13}}$ $= \frac{10\sqrt{13}}{13}$	2 marks – correct solution 1 mark – correct substitution into formula

b	$T_n = ar^{n-1}$ $\frac{T_8}{T_5} = \frac{ar^7}{ar^4} = r^3 = \left(\frac{1}{243}\right) \div \frac{1}{9}$ $r^3 = \frac{1}{27}$ $r = \frac{1}{3}$	2 marks – correct solution. 1 mark – determining expression for r^3
c	$\sum_{m=1}^{\infty} 5 \left(\frac{2}{5}\right)^{m-1}$ - expression is limiting sum therefore $T_1 = a = 5 \left(\frac{2}{5}\right)^0 = 5$ $r = \frac{2}{5}$ $S_{\infty} = \frac{a}{1-r}$ $= \frac{5}{1-\frac{2}{5}}$ $= \frac{5}{\frac{3}{5}}$ $= \frac{25}{3}$	2 marks – correct solution 1 mark – correct first term and common ratio identified. -
d	$\int \frac{1}{(2x+3)^5} dx$ $= \int (2x+3)^{-5} dx$ $= \frac{(2x+3)^{-4}}{2 \times -4} + C$ $= -\frac{1}{8(2x+3)^4} + C$	2marks – correct solution 1 mark – incorrect denominator

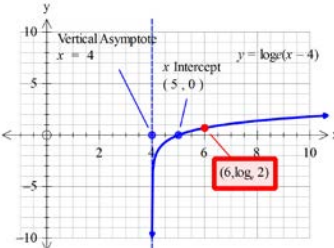
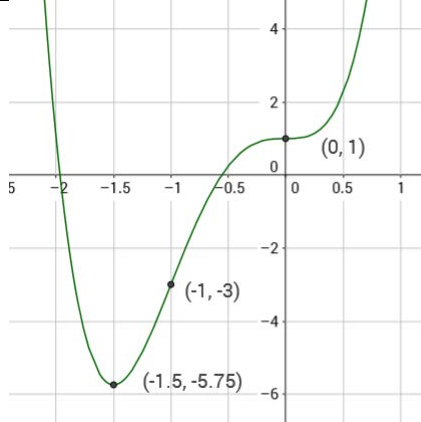
e	 $\angle BEA = 180 - 100 = 80$$\angle CBE = \angle BEA + \angle BAE$$3x = 80 + x$$2x = 80$$x = 40$ straight angle ext $\angle$ of $\Delta = \Sigma$ opp interior $\angle$	2 marks – correct answer with reasons 1 mark – correct answer with incomplete reasons												
f-i	<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>$y = \frac{10\sqrt{x}}{x+5}$</td><td>0</td><td>$\frac{10}{6} = \frac{5}{3}$</td><td>$\frac{10\sqrt{2}}{7}$</td><td>$\frac{10\sqrt{3}}{8} = \frac{5\sqrt{3}}{4}$</td><td>$\frac{20}{9}$</td></tr></table> $A \cong \frac{h}{2} \{ y_0 + 2(y_1 + y_2 + y_3 + y_4) \}$$\cong \frac{1}{2} \times \left(0 + 2 \left(\frac{5}{3} + \frac{10\sqrt{2}}{7} + \frac{5\sqrt{3}}{4} \right) + \frac{20}{9} \right)$$\cong 6.96$	x	0	1	2	3	4	$y = \frac{10\sqrt{x}}{x+5}$	0	$\frac{10}{6} = \frac{5}{3}$	$\frac{10\sqrt{2}}{7}$	$\frac{10\sqrt{3}}{8} = \frac{5\sqrt{3}}{4}$	$\frac{20}{9}$	2 marks – correct solution 1 mark – correct table or similar
x	0	1	2	3	4									
$y = \frac{10\sqrt{x}}{x+5}$	0	$\frac{10}{6} = \frac{5}{3}$	$\frac{10\sqrt{2}}{7}$	$\frac{10\sqrt{3}}{8} = \frac{5\sqrt{3}}{4}$	$\frac{20}{9}$									
f-ii	 As the curve is concave down, the trapeziums would lie under the curve hence estimate is an underestimate.	1 mark – correct answer with reasons.												

Markers Comments.

12-a-i a number of students incorrectly interpreted as midpoint

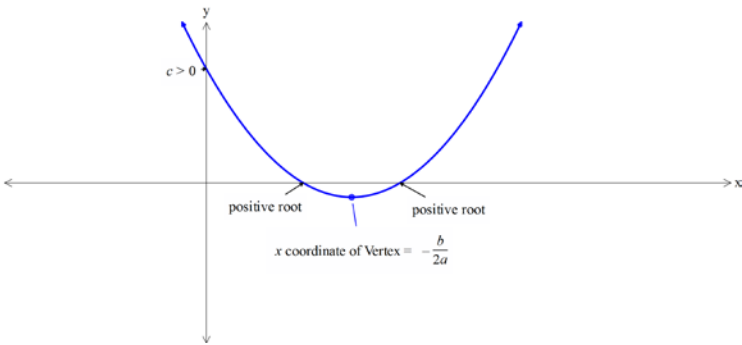
Q13a	$\int (1 + \tan x) dx = \int \left(1 + \frac{\sin x}{\cos x} \right) dx$ $= x - \ln \cos x + c$	2 marks correct solution from correct working 1 mark correct $\tan x = \frac{\sin x}{\cos x}$
Q13b-i	Distance from P = 150 - distance travelled $= 150 - 50t$	1 mark – correct demonstration.
Q13b-ii	$d = \sqrt{(150 - 50t)^2 + (40x)^2}$ $= \sqrt{22500 - 2 \times 150 \times 50t + 2500t^2 + 1600t^2}$ $= \sqrt{4100t^2 - 15000t + 22500}$	2 marks correct solution from correct working 1 mark correct substitution into Pythagoras' Theorem
13b-iii	As $4100t^2 - 15000t + 22500$ is a parabola with $a = 4100$, the parabola is concave up and therefore a minimum. $f'(t) = 8200t - 15000$ stationary point at $f'(t) = 0$ $0 = 8200t - 15000$ $t = \frac{15000}{8200}$ $= 1.82926.....$ $= 1.8$	2 marks correct solution from correct working. 1 marks correct x with no justification for minimum value
13c	$y = 4 \tan(2x) - \frac{4x^2}{\pi}$ $y' = 4 \times \sec^2(2x) \times 2 - \frac{8x}{\pi}$ $= 8 \left(\sec^2(2x) - \frac{x}{\pi} \right)$ at $x = \frac{\pi}{12}$ $y' = 8 \left(\sec^2\left(\frac{2\pi}{12}\right) - \frac{\frac{\pi}{12}}{\pi} \right)$ $= 8 \left(\sec^2\left(\frac{\pi}{6}\right) - \frac{1}{12} \right)$ $= 8 \left(\left(\frac{2}{\sqrt{3}}\right)^2 - \frac{1}{12} \right)$ $= 8 \left(\frac{4}{3} - \frac{1}{12} \right)$ $= 10$	2 marks correct solution from correct working. 1 mark correct derivative.

13di	ZUVWXY regular hexagon (given) $\text{Interior angle} = \frac{180(6-2)}{6}$ $= 120^\circ$ $\angle AUV = \angle AVU$ $= 180^\circ - 120^\circ \text{ (straight } \angle \text{)}$ $= 60^\circ$ $\angle VAU = 180^\circ - (60^\circ + 60^\circ) \text{ (} \angle \text{ sum of } \Delta \text{)}$ $= 60^\circ$ $\therefore \Delta AVU \text{ is equilateral}$	2 marks correct solution from correct working with reasons. 1 mark finding an interior angle of hexagon
13dii	$\angle ZAW = \angle ZFW$ $= 60^\circ \text{ (equilateral } \Delta s \text{)}$ $\angle AZF = \angle AWF$ $= 120^\circ \text{ (} \angle \text{ of regular hexagon)}$ $\therefore ZAWF \text{ is a parallelogram (opposite } \angle = \text{)}$ $AU = AV \text{ (} \Delta AVU \text{ equilateral)}$ $ZU = VW \text{ (given)}$ $\therefore ZA = WA$ $\therefore ZAWF \text{ is a rhombus}$ $\text{(parallelogram with a pair of adjacent sides =)}$	2 marks correct solution from correct working with reasons. 1 mark proving ZAWF is a parallelogram
13e	$\frac{d}{dx} \left(\frac{\cos x}{1 - \sin x} \right)$ $u = \cos x \quad v = 1 - \sin x$ $u' = -\sin x \quad v' = -\cos x$ $\frac{d}{dx} \left(\frac{\cos x}{1 - \sin x} \right) = \frac{-\sin x(1 - \sin x) - \cos x(-\cos x)}{(1 - \sin x)^2}$ $= \frac{-\sin x + \sin^2 x + \cos^2 x}{(1 - \sin x)^2}$ $= \frac{1 - \sin x}{(1 - \sin x)^2}$ $= \frac{1}{1 - \sin x}$	2 marks correct solution from correct working 1 mark correct u' and v'

Q14a)		2 marks correct answer with 2nd pt 1 mark correct asymptote and x-int																		
bi)	$y' = 16x^3 + 24x^2$ $y' = 0 \text{ for a S.P.}$ $8x^2(2x + 3) = 0$ $x = 0 \text{ and } x = -\frac{3}{2}$ $y = 1 \text{ and } y = -\frac{23}{4} \text{ respectively}$ Using a slope table <table border="1" data-bbox="292 795 940 985"><tr><td>x</td><td>-2</td><td>-1.5</td><td>-1</td><td>0</td><td>1</td></tr><tr><td>y'</td><td>-32</td><td>0</td><td>8</td><td>0</td><td>40</td></tr><tr><td>slope</td><td>\</td><td>-</td><td>/</td><td>-</td><td>/</td></tr></table> $(-\frac{3}{2}, -\frac{23}{4})$ is a minimum and $(0,1)$ a horizontal point of inflexion Or use of y'' to show concavity change $y'' = 0$ is not sufficient for a point of inflection	x	-2	-1.5	-1	0	1	y'	-32	0	8	0	40	slope	\	-	/	-	/	4 marks: correct answer 3 marks: correct coordinates and their nature, showing HPOI 2 marks: correct co-ords 1 mark: correct first derivative
x	-2	-1.5	-1	0	1															
y'	-32	0	8	0	40															
slope	\	-	/	-	/															
ii)	$y'' = 48x^2 + 48x$ $y'' = 0 \text{ for possible points of inflexion}$ $48x(x + 1) = 0$ $x = -1, 0$ or sub $x=1$ into 2nd derivative this point of inflection is NOT a horizontal point of inflection	2 marks: correct answer 1 mark: correct second derivative																		
iii)		3 marks: correct answer 2 marks: correct shape 1 mark correct stationary points Cfe applicable NOTE some very poor graphs were presented																		

c)i	$x = 70e^0 - 20(0)$ $x = 70\text{m}$	1 mark correct answer
ii)	$= -7e^{\frac{t}{10}} - 20$ $= 0$ for particle at rest $e^{-\frac{t}{10}} \neq -\frac{20}{7}$ since $e^{-\frac{t}{10}} > 0$ for all	
iii)	from i) $t=0$ and $x=70$ $t=3$ $x=-8.14$ therefore distance travelled is 78.14m	1 mark correct answer

Q15 a	$f''(x) = 12x - 10$ $f'(x) = 6x^2 - 10x + c_1$ stationary points at $f'(x) = 0$ when $x = 2$ $f'(2) = 6(2)^2 - 10(2) + c_1 = 0 \Rightarrow c_1 = -4$ $\therefore f'(x) = 6x^2 - 10x - 4$ $f(x) = 2x^3 - 5x^2 - 4x + c_2$ sub point $(2, -10)$ $-10 = 2(2)^3 - 5(2)^2 - 4(2) + c_2 \Rightarrow c_2 = 2$ $\therefore f(x) = 2x^3 - 5x^2 - 4x + 2$	3 marks correct solution from correct working 2 marks- partial correct/ one error 1 mark- partial correct with one required process
Q15 b	Method 1 $\int_{-2}^2 x^7 - x + k \, dx = \left[\frac{x^8}{8} - \frac{x^2}{2} + kx \right]_{-2}^2$ $= \left(\frac{256}{8} - \frac{4}{2} + 2k \right) - \left(\frac{256}{8} - \frac{4}{2} - 2k \right)$ $= 4k$ $\therefore \begin{aligned} 4k &= 16 \\ k &= 4 \end{aligned}$ Method 2 $\int_{-2}^2 (x^7 - x + k) \, dx$ $= \int_{-2}^2 (x^7 - x) \, dx + \int_{-2}^2 k \, dx$ $= 0 + [kx]_{-2}^2$ (as first function is an odd function) $\therefore 2k - (-2k) = 16$ $4k = 16$ $k = 4$	2 marks correct solution from correct working 1 mark for integration
Q15 c(i)	when $t = 0$ $C(0) = 105$ when $t = 1$ $C(1) = 0.6 \times 105 = 63$ and $C(1) = 105e^{-k}$ $\therefore 105e^{-k} = 63$ $e^{-k} = \frac{63}{105}$ $-k = \ln\left(\frac{63}{105}\right)$ $k = -\ln(0.6)$	2 marks correct solution from correct working 1 mark for using 0.4 in equation and solving with correct process Common error: Using 40% instead of 60%

Q15 c(ii)	$105e^{-kt} = 10$ $e^{-kt} = \frac{10}{105} = \frac{2}{21}$ $-kt = \ln\left(\frac{2}{21}\right)$ $t = \ln\left(\frac{2}{21}\right) \div \ln(0.6)$ $t \approx 4.6 \text{ hours (2 sigfig)}$	1 marks correct solution from correct working 1 mark if error carried from part i) showing correct process Common error: Rounding to 3 sig fig (4.60)
Q15 d(i)	For 2 real roots $\Delta > 0$ $\Delta = b^2 - 4ac$ $= (-2k)^2 - 4(2k + 3)$ $= 4k^2 - 8k - 12$ $4k^2 - 8k - 12 > 0$ $k^2 - 2k - 3 > 0$ $(k + 1)(k - 3) > 0$ $\therefore k < -1 \text{ or } k > 3$	2 marks correct solution from correct working 1 mark for discriminant
Q15 d(ii)	Method 1 For there to be two roots, k must satisfy $k < -1$ or $k > 3$ Since $a > 0$, parabola is concave up, for there to be two positive roots k must satisfy additional conditions of constant term > 0 and $-\frac{b}{2a} > 0$ - see diagram below.  $\therefore c > 0$ i.e. $2k + 3 > 0$ $k > -\frac{3}{2}$ Also, axis of symmetry $x = -\frac{b}{2a}$ must be > 0 $x = -\frac{b}{2a} = \frac{2k}{2(1)} = k$ $\therefore k > 0$ Therefore to satisfy all conditions, k must simultaneously satisfy	2 marks correct solution from correct working 1 mark for considering one condition Common error: Not finding the other condition of $k > 0$ Using the quadratic equation and then solving for $x > 0$ resulted in algebraic error (If no error, successfully found $k > -3/2$ condition but failed to find $k > 0$ condition)

	$k > -\frac{3}{2}, k > 0 \text{ and } k > 3$ therefore $k > 3$ satisfies all conditions. Method 2 $\alpha + \beta > 0 \text{ and } \alpha \beta > 0$ $\alpha + \beta = -\frac{b}{a} = 2k$ $\therefore 2k > 0 \Rightarrow k > 0$ $\alpha \beta = \frac{c}{a} = 2k + 3$ $\therefore 2k + 3 > 0 \Rightarrow k > -\frac{3}{2}$ Considering all conditions $k > 3$	
Q15 e	Method 1 $h = \int_0^5 110(t+4)^{-2} dt + 0.5$ $= 110 \left[\frac{(t+4)^{-1}}{-1} \right]_0^5 + 0.5$ $= 110 \left[-\frac{1}{t+4} \right]_0^5 + 0.5$ $= 110 \left[-\frac{1}{9} + \frac{1}{4} \right] + 0.5$ $\approx 15.8m$ Method 2 $h = \int 110(t+4)^{-2} dt$ $= -\frac{110}{t+4} + c$ when $t = 0, h = 0.5$ $0.5 = -\frac{110}{4} + c$ $c = 28$ $\therefore h = -\frac{110}{t+4} + 28$ when $t = 5$ $h = -\frac{110}{9} + 28 \approx 15.8$	3 marks correct solution from correct working 2 marks for definite integral but omitting the 0.5 1 mark for integration 3 marks correct solution from correct working 2 marks for integration and constant 1 mark for integration

Question 16		
16 a i)	$P = \$ 250\,000, R = 1.004$ $A_n = A_{n-1}R - M$ $\Rightarrow A_1 = 250\,000(1.004) - M$ $A_2 = A_1R - M$ $\Rightarrow A_2 = [250\,000(1.004) - M](1.004) - M$ $A_2 = 250\,000(1.004)^2 - 1.004M - M$ $A_2 = 250\,000(1.004)^2 - M(1 + 1.004)$	2 Marks: Correct solution 1 Mark: correct expression for A_1
16 a ii)	$A_n = 250\,000(1.004)^n - M(1 + 1.004 + \dots + 1.004^{n-1})$ $S_n = a \left[\frac{R^n - 1}{R - 1} \right] \text{ and } a = 1, R = 1.004$ $= 250\,000(1.004)^n - M \left[\frac{1.004^n - 1}{0.004} \right]$	1 Mark: Correct solution Must show series up to and including 1.004^{n-1}
16 a iii)	$A_{120} = 250\,000(1.004)^{120} - M \left[\frac{1.004^{120} - 1}{0.004} \right]$ $0 = 250\,000(1.004)^{120} - M(153.631\dots)$ $M = \frac{250\,000(1.004)^{120}}{153.631\dots}$ $M = 2627.2655\dots \approx \2627	1 Mark: Correct solution
16 a iv)	$A_n = 250\,000(1.004)^n - 3500 \left[\frac{1.004^n - 1}{0.004} \right]$ $0 = 250\,000(1.004)^n - 875\,000(1.004^n - 1)$ $0 = 250\,000(1.004)^n - 875\,000(1.004)^n + 875\,000$ $0 = -625\,000(1.004)^n + 875\,000$ $(1.004)^n = \frac{875\,000}{625\,000}$ $n \log_e(1.004) = \log_e \left(\frac{875}{625} \right)$ $n = \log_e \left(\frac{875}{625} \right) \div \log_e(1.004) = 84.286\dots$ $\therefore$ 84 full payments and part payment of $0.286\dots(\\$3500) = \\$ 1001.64$	3 Marks: Correct solution. 2 Marks: Correct derivation of $n = 84.286\dots$ 1 Mark:
16 b i)	$\frac{\sec\theta}{\csc\theta} = \left(\frac{1}{\cos\theta} \right) \div \left(\frac{1}{\sin\theta} \right) = \frac{\sin\theta}{\cos\theta} = \tan\theta$	1 Mark: Correct solution

16 b ii)	$3\sec^2\left(\frac{x}{2}\right) = \operatorname{cosec}^2\left(\frac{x}{2}\right) \Rightarrow \frac{3}{\cos^2\left(\frac{x}{2}\right)} = \frac{1}{\sin^2\left(\frac{x}{2}\right)}$ $\frac{\sin^2\left(\frac{x}{2}\right)}{\cos^2\left(\frac{x}{2}\right)} = \frac{1}{3} \Rightarrow \tan^2\left(\frac{x}{2}\right) = \frac{1}{3}$ $\tan\left(\frac{x}{2}\right) = \pm \frac{1}{\sqrt{3}}$ $0 \leq x \leq 2\pi$ $\Rightarrow 0 \leq \frac{x}{2} \leq \pi \Rightarrow \frac{x}{2} = \frac{\pi}{6}, \frac{5\pi}{6} \Rightarrow x = \frac{\pi}{3}, \frac{5\pi}{3}$	
16 c i)	$x = \frac{e}{3} \rightarrow y = \log_e(3x) \Rightarrow y = \log_e\left(3 \cdot \frac{e}{3}\right) = \log_e(e) = 1$	
16 c ii)	$V_1 = \text{cylinder} = \pi r^2 h = \pi \left(\frac{e}{3}\right)^2 \cdot 1 = \frac{\pi e^2}{9}$ $V_2 = \text{solid of revolution} = \pi \int_0^1 x^2 dy$ $y = \log_e(3x) \rightarrow 3x = e^y \rightarrow x^2 = \left(\frac{e^y}{3}\right)^2 = \frac{e^{2y}}{9}$ $V_2 = \left(\frac{\pi}{9}\right) \int_0^1 e^{2y} dy = \frac{\pi}{9} \left[\frac{1}{2} e^{2y} \right]_0^1 = \frac{\pi}{18} [e^2 - 1]$ $V = V_1 - V_2 = \frac{\pi e^2}{9} - \frac{\pi}{18} [e^2 - 1]$ $V = \frac{\pi(e^2 + 1)}{18}$	